



Emulium® Dolcea MB

Soft by nature



People make our name



The next generation of natural cosmetics



It is not news that natural skincare has gone mainstream but consumer requirements have further increased. It is no longer enough to be just naturally derived; cosmetics today should be biodegradable, sustainably and responsibly sourced, certifiable, vegan and perfectly stable over a large range of temperatures.

Formulating stable and sensorially pleasing cosmetics using only natural and biodegradable ingredients is not an easy task. In particular, natural gelling agents, mostly polysaccharides, are less stabilizing than synthetic gels, especially under heat stress. The path to emulsion stability is difficult and time-consuming and involves discovery of an optimal combination of oil and aqueous based thickeners. Yet formulators are increasingly challenged by short development times.

From this observation, Gattefossé has developed Emulium® Dolcea MB, the most robust of its range of natural O/W emulsifiers. Its composition has been finely tuned to provide the best performance/stability/sensory ratio. It is easy to use, quick to emulsify and stabilizes within minutes. It forms soft, highly moisturizing textures that are guaranteed to delight the consumer.



A fine-tuned composition

Every component of Emulium® Dolcea MB plays a critical role to ensure stability and impact to the final emulsion:

EMULSIFICATION & STABILIZATION

Sodium Stearoyl Glutamate (SSG) is used as a strong primary emulsification agent.

Polyglycerol-3 esters of sunflower and jojoba waxes, composed of a large hydrophilic head and a long hydrophobic tail, ensure steric repulsion between droplets and bring good heat stability.

SUPPORT

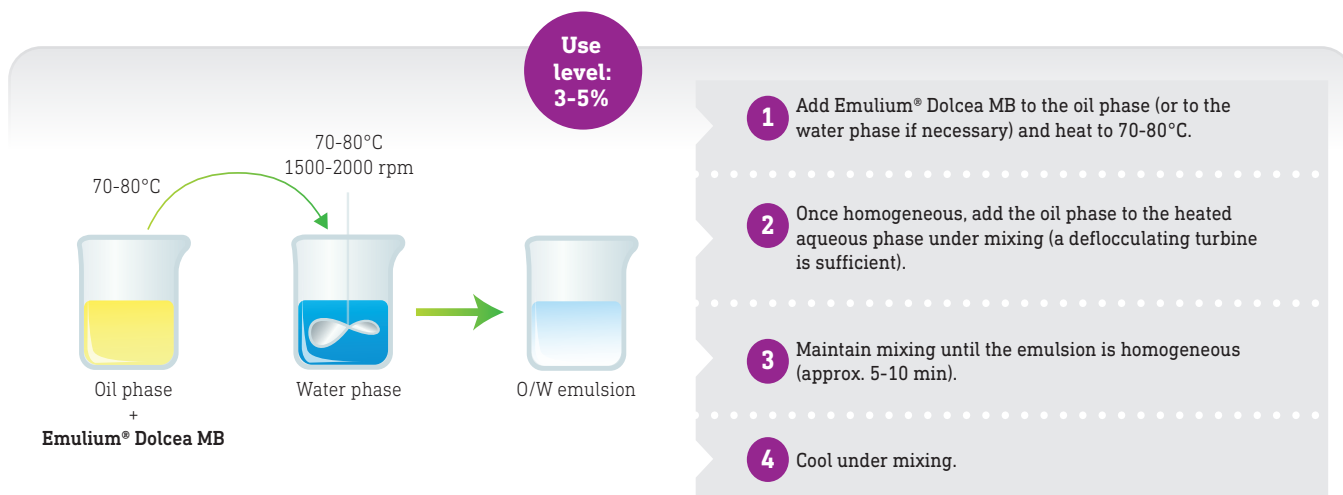
Glyceryl stearate and cetearyl alcohol migrate to the interface, finalizing the robust structure. Cetearyl alcohol also hydrates in the water phase, stabilizing via the formation of lamellar phases.

SENSORY & MOISTURIZING PROPERTIES

Sunflower and jojoba waxes are transesterified to create novel lipophilic esters. In the oil phase, they bring a characteristic softness during application and after absorption, and immediate and long-term moisturizing properties.

The most robust of our natural emulsifiers

As a result, Emulium® Dolce MB is capable of emulsifying and stabilizing emulsions with a high amount of oil phase (up to 40%). In the form of pellets, it is very easy to formulate, does not require high shear and creates very fine emulsions. Flexible, it can be incorporated in the oil phase and in the water phase and can be used in a concentrated emulsion process to save energy and time.

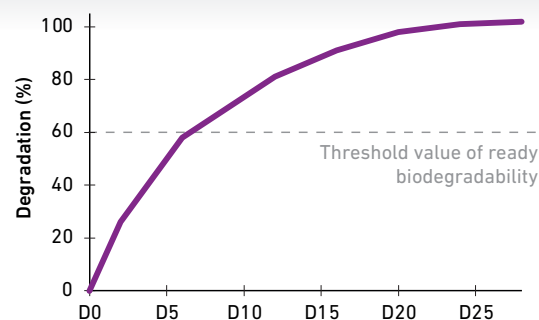


Compatibility with cosmetic ingredients – emollients, polyols, preservative systems, stabilizers, organic UV filters, TiO₂ – is excellent. It is also compatible with alcohol allowing the formulation of perfume ancillaries. It performs with a wide range of natural gelling agents that normally present limited stabilizing capacity:

- Xanthan gum
- Cellulose derivatives
- Starches
- Sclerotium gum
- Alginates

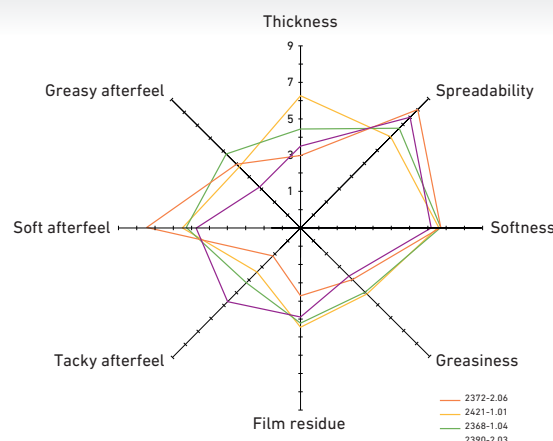
Biodegradability⁽¹⁾

The composition of Emulium® Dolce MB has been carefully chosen to ensure excellent biodegradability. Results showed that 100% of Emulium® Dolce MB, in equivalent organic carbon, has been biodegraded after only 28 days.



Extreme softness

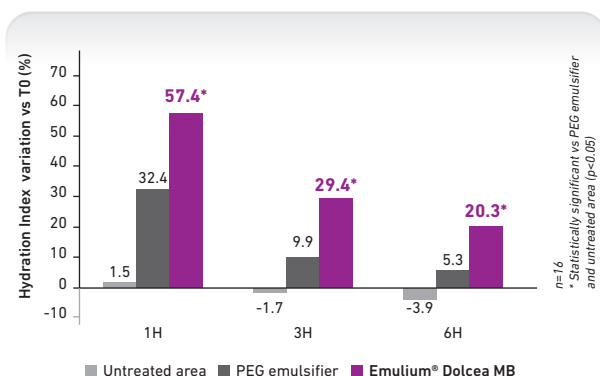
Sensory analysis, carried out on over ten formulations of highly varied composition in oil phase and polysaccharide nature, has shown that the sensory personality of **Emulium® Dolce MB is characterized by an extreme softness in particular on skin afterfeel**. This softness can be attributed to the mixed esters of jojoba and sunflower which combine spreadability and body. This effect is impossible to obtain by simple blending alone.



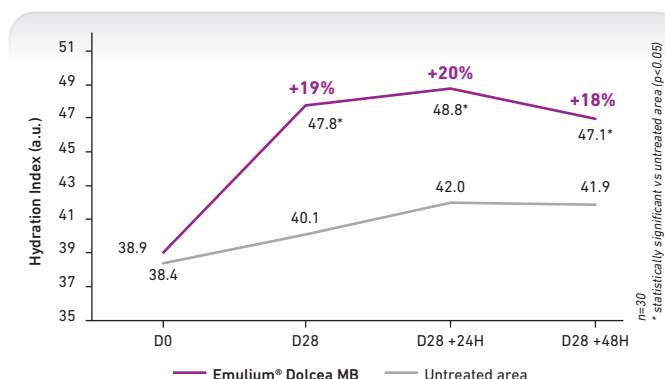
⁽¹⁾ Ready biodegradability test (OCDE 301 F)

48H moisturization

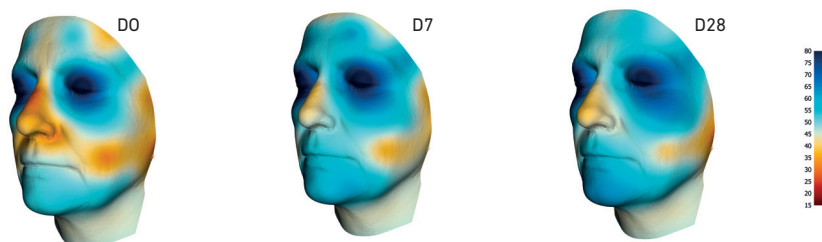
A corneometry test has been performed to highlight the moisturizing effect of Emulium® Dolcea MB, compared to a PEG emulsifier (both used at 4% in a simple emulsion) and versus an untreated area⁽²⁾.



A natural and simple emulsion containing 4% of Emulium® Dolcea MB was tested on women with dry skin⁽³⁾. Skin moisture measurements were performed before the first application, after 28 days, 24h and 48h after the last application to assess skin moisture remanence.



Hydration mapping⁽⁴⁾



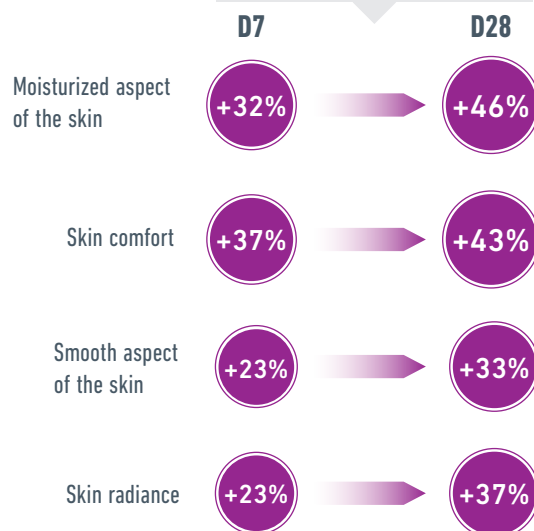
Emulium® Dolcea MB is the choice emulsifier for long-lasting moisturizing cosmetics. The skin hydration level is homogeneously improved over the whole face.

Acclaimed by consumers⁽⁵⁾

JUST AFTER THE FIRST APPLICATION



ON BARE SKIN



The moisturized and smooth aspect of the panelists skin was improved. The skin feels comfortable and is visibly radiant.

⁽²⁾ Corneometry test performed on 8 volunteers (n=16 forearms) during 6 hours

⁽³⁾ Clinical evaluation under dermatological control on 33 volunteers (30-62 years old), twice-a-day application during 28 days

⁽⁴⁾ Hydration mapping performed on 23 panelists to visualize the hydration index on a mean face obtained by fringe projection

⁽⁵⁾ Self-assessment performed on 33 panelists after the first application, at D7 and D28



Emulium® Dolcea MB

is a PEG-free O/W emulsifier of 100% natural origin, easy to formulate and flexible, which makes it possible to formulate from fluid textures to thick butters.

Emulium® Dolcea MB shows high compatibility with cosmetic ingredients and provides excellent stability alongside polysaccharide gelling agents.

With immediate and long-lasting moisturizing capacity, this ingredient brings functionality and efficacy to a wide range of applications.

Featuring a strong sensory signature, it brings extreme softness to cosmetic emulsions. COSMOS approved, Emulium® Dolcea MB can be used in organically certified, vegan and readily biodegradable formulations.

Characteristics

- O/W emulsifier (pellets)
- PEG-free, 100% natural origin
- HLB = 10

Applications

- From fluid to thick textures
- Skin care
- Color cosmetics
- Sun care
- Organically certified, vegan and readily biodegradable formulations

Use level

3-5%

INCI name: Cetearyl Alcohol (and) Glyceryl Stearate (and) Jojoba Esters (and) Helianthus Annuus (Sunflower) Seed Wax (and) Sodium Stearoyl Glutamate (and) Water (and) Polyglycerin-3

Worldwide approval for cosmetic applications

Listed on IECIC (Inventory of Existing Cosmetic Ingredients in China) 2015

COSMOS approved

Readily biodegradable

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www.gattefosse.com



Corporate Headquarters

36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - **France**
+(33) 4 72 22 98 00